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Fairness in Recommender Systems: Evaluation Approaches and Assurance Strategies

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With the wide application of recommender systems, the potential impacts of recommender systems on customers, item providers and other parties have attracted increasing attention. Fairness, which is the quality of treating people equally, is also becoming important in recommender system evaluation and algorithm design. Therefore, in the past years, there has been a growing interest in fairness measurement and assurance in recommender systems. Although there are several reviews on related topics, such as fairness in machine learning and debias in recommender systems, they do not present a systematic view on fairness in recommender systems, which is context aware and has a multi-sided meaning. Therefore, in this review, the concept of fairness is discussed in detail in the various contexts of recommender systems. Specifically, a comprehensive framework to classify fairness metrics is proposed from four dimensions, i.e., *Fairness for Whom*, *Demographic Unit*, *Time Frame*, and *Quantification Method*. Then the strategies for eliminating unfairness in recommendations, fairness in different recommendation tasks and datasets are reviewed and summarized. Finally, the challenges and future work are discussed.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Information systems** → **Recommender systems**;

Additional Key Words and Phrases: Recommender system, fairness, survey

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1 INTRODUCTION

Modern recommender systems learn users preference from massive amounts of data and provide users with recommendation results that can match their preferences and satisfy other goals. The past few years have witnessed the great success of recommender systems in areas ranging from e-commerce, online media, and online travel services to investment decisions.

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How to evaluate a recommender system is an important issue for their practical application. Generally, a recommender system is evaluated in terms of the accuracy of the recommendation results [45], that is, the proportion of items that a user clicks on or selects from the recommended ones. However, only focusing on recommendation accuracy will narrow the users' vision, causing users to enter so-called information cocoons [152], or it may cause the market environment to be affected by dominant groups, creating a vicious circle in which the rich get richer and the poor poorer [31]. Therefore, more metrics should be taken into consideration, such as the diversity of results [2, 14, 100], novelty [52], and so on. Of these, the fairness of recommendation is an important metric that is attracting increasing attention.

Modern recommender systems suffer from the serious problem of unfairness. For example, in the MovieLens dataset [48], the most used dataset in recommendation research, there are more ratings from male than female users, and therefore, the recommendations will deviate from female users' preferences, i.e., the quality of recommendations to female and male users is unfair [53]. As another example, in the item list of recommendations, lower-ranked items often receive disproportionately less attention than they deserve according to ranking relevance. The unfair attention [59] brought by position on the web page results in a rich-get-richer problem and becomes a barrier for unpopular services to startup. Recently, the issue of fairness in recommendation has been investigated in many domains, such as *Music* recommendation [84], *Job* recommendation [61], and *Book* recommendation [33].

Fairness is also becoming an important issue in machine learning, and a few papers have recently reviewed fairness issues relating to machine learning. For example, fairness notions in machine learning are listed in Reference [79] and the bias in the data that leads to unfairness is analyzed in References [82, 87]. Moreover, a taxonomy for fairness definitions has been created by machine learning researchers to avoid the existing bias in AI systems. Methods to address the unfairness problems are reviewed in Reference [82], and Reference [18] provides an overview of approaches to mitigate (social) biases and increase fairness in the machine learning literature. Because recommender systems have other optimization goals besides accuracy and the implications of fairness in recommender systems are different from machine learning, reviews are also published on the topic of fairness in recommendations. For example, Reference [116] provides an overview of the definitions of fairness in ranking and recommendation in various settings. Reference [93] introduces the concept of fairness in recommendations and reviews the methods for achieving fairness. In addition, Reference [21] provides a comprehensive review on bias and debias in recommender systems. In these reviews, the types of fairness are often divided into customer fairness, provider fairness, and side stakeholder fairness.

Although these reviews try to summarize the current research on the fairness in recommender systems, they are not sufficient in terms of their scope and depth. First, they do not present a comprehensive discussion on the implications of fairness in recommender systems. There are quite a few terms that have similar meanings to fairness, such as *justice* [88], *equality* [19, 133], *equity* [140], and *debiasing* [21]. This leads to the fact that there are many different understandings and definitions of fairness in the literature. Second, fairness in recommendation is closely related to contexts [3, 99, 128], which means the meanings of fairness should be contextualized by different applications. This makes fairness in recommender systems more complex than it is in general machine learning areas. Current reviews do not examine the topic of fairness in recommender systems by considering substantial application contexts.

This article provides a more systematic review on the topic of fairness in recommender systems. The context of recommender system is introduced first to define the background of fairness. Then we discuss the fairness issue from three layers, i.e., the concept layer, the fairness notion layer, and the fairness metrics layer. The conceptual meaning of fairness is discussed based on a

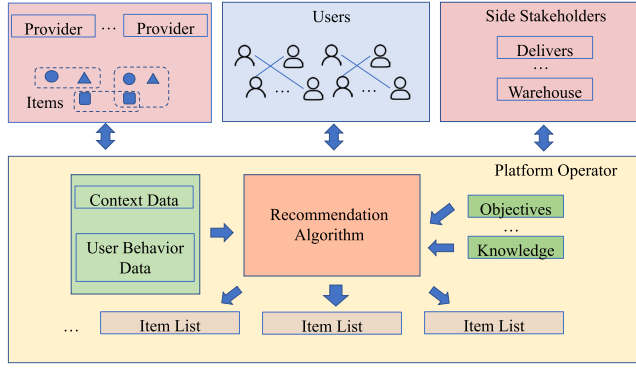


Fig. 1. Recommender system and its environment.

comparison with other related concepts. Then the fairness notions that are applied in recommendation are listed. The introduction and discussions on the concept and notions of fairness in recommendation lays a foundation for the fairness measurement, for which a four-dimensional framework is proposed to accommodate diversified measurement definitions in the literature. Moreover, we list the desirable properties of fairness metrics, since it is not possible to define a unified metrics. For the fairness assurance strategies, although we also classify them into pre-processing, in-processing, and post-processing ones as some reviews do, we give more detailed strategy classifications in each category. We also present a discussion on fairness in different recommendation tasks. Finally, we point out the challenges and future work on this topic.

The article is organized as follows. Section 2 discusses the concept of fairness in recommender systems. Section 3 classifies the measurement of fairness in recommender systems in detail. We introduce the strategies for fair recommendation in Section 4. The potential topics of fairness in different recommendation tasks are discussed in Section 5. We detail the datasets used in the research on fairness in recommender systems in Section 6. In Section 7, we point out some challenges and make suggestions for future work. Finally, we conclude the article in Section 8.

2 FAIRNESS IN RECOMMENDER SYSTEMS

2.1 Recommender Systems in a Business Environment

A recommender system is usually deployed in a business environment. Therefore, to evaluate the impact of the recommender system, the interests of all the stakeholders in its business environment should be considered.

A recommender system and its business environment are shown in Figure 1. Generally, a recommender system is a component of a platform that is developed and maintained by a platform operator. In some cases, the functions of recommender systems can also be provided as a service to the platform operator by the third party.

Users (such as customers, visitors, app users or readers) are the center of recommender systems. Users who receive recommendations have attributes (including their social relationships) that could have been provided when they registered with the system or were perhaps obtained from other channels.

Because recommender systems have been applied to an increasing number of areas, objects (such as products, movies, air tickets, or articles) that are recommended become diversified, and even actions can be recommended [137]. In this article, we call these *items* for expression

convenience. Items also have attributes and can be classified into types. Items are owned or sold by providers who want to ensure that more of their items are accepted by users.

Platform operators have different revenue models, including an advertising revenue model, subscription revenue model, transaction fee revenue model, sales revenue model, or affiliate revenue model. In spite of the different revenue models, the number of users and their access are the key. This is why recommender systems have become an indispensable function on all platforms.

However, there are other participants (*side stakeholders*) in the business environment on which recommender systems have an impact. For example, a sharp increase in a specific product's sales caused by recommendations will result in transport pressures for some logistics companies.

In recommender systems, *items* are recommended based on the items' information, user's preferences and context data. During this process, different forms of *knowledge* (such as knowledge maps or rules) may also be utilized to optimize the recommendation results. The recommendations can be personal or non-personal. For non-personal recommendation, the algorithm does not take into account the personal preferences of the users and the recommendations produced are identical for each customer. The recommendations can either be manually selected, based on the popularity of items or the recommendations can be the top-N new products.

For personal recommendation, algorithms try to learn users' preferences according to their attributes and behavior data. The earliest widely used personalized recommendation algorithm was collaborative filtering [43]. The basic collaborative filtering approach is a neighborhood-based model, which recommends items preferred by users that are similar to the current user [51] or items that are similar to the current user's selected items [74].

In recent years, latent factor based models have become the mainstream of the collaborative filtering approaches, because they can deal with the sparsity problem in recommendations. User latent features and item latent features can be learned from the matrix factorization algorithm [60] or tensor factorization algorithm. Then, the inner product of the user's latent feature vector and an item's latent feature factor represent the preference of the user for the item.

The content-based model [92] can comprehensively utilize various information of users, items, and context to generate recommendation results, which can help solve cold-start problems. A logistic regression model can be used to predict whether a user likes an item or not based on the features of users and items. However, the logistic regression model does not consider feature combinations. Therefore, improved algorithms such as factorization machines [101], field-aware factorization machines [54], large scale piecewise linear model [37], and so on, are proposed. A content-based model and collaborative filtering can also be used in a hybrid way.

With the development of deep learning algorithms, deep learning models [62] have also been widely used in recommendation systems. **Neural collaborative filtering (NCF)** [50] replaces the inner products in MF with a deep neural network that learns the user-item interactions.

In combination with other advanced neural network structures, such as reinforcement learning neural networks [143], adversarial learning neural networks [145], graph neural networks [20, 129, 136], and so on, deep learning recommendation models have become the mainstream in the recommendation field.

In the past few years, the advancement of recommendation techniques have been mainly driven by the motivation to improve recommendation accuracy, which can be regarded as a typical machine learning problem. However, in the recommendations and related processes, each participant has interests that should be respected. For example, for users, their interests are recommended items that should be relevant to their needs and preferences; for item providers, their interests are items that should be shown to users with an equal chance. This requires the design of recommendation algorithms that need to care not only about users interests but also other parties'

interests. Therefore, fairness in recommendations should also be considered in a more complex context.

2.2 The Concept of Fairness

Generally speaking, fairness is the quality of treating people equally or in a way that is reasonable according to the definition in the Oxford dictionary. This definition is casual. Therefore, we compare fairness with related terms to clarify its meanings.

Bias [27] is an objective property of the dataset or the algorithm, which is an inclination toward something. Usually bias can be detected numerically. In contrast, fairness is a social construct, and it does not exist in objective reality but is created and accepted by people. The term “*debiasing*” has also appeared in several related papers. The work related to debiasing and the work on fairness will overlap in content, but there are also parts that differ from each other. Debiasing is a passive behavior from the point of view of purpose, and there are clear goals and metrics. But fairness is a very subjective state. Different people have different definitions and understandings of fairness from different angles. We can say that eliminating bias can achieve demographic fairness, because such fairness standards are objective, but other fairness notions may not be achieved. At the same time, conversely speaking, achieving fairness does not necessarily eliminate bias.

Equality [4] is a status where each individual or group is given the same amount of resources, attention or outcome. *Equity* [72] is also a status where each individual or group is given the resources they need to succeed. Equality and equity can be one approach to achieve fairness.

Justice [8] is another related concept, whose meaning is very broad with room for many perspectives including ethics and legal and political philosophy. Broadly speaking, justice is often seen as a quality of being just or fair. In many places, justice and fairness are used interchangeably. But justice can also be denoted as a process, or an action. In contrast, fairness only refers to the result or a status.

The subtle difference between the meanings of these concepts mean that these words are often used without being distinguished in the literature on machine learning including recommender systems.

2.3 Fairness Notions for Recommendation

To make the concept of fairness more tangible in practice, fairness notions are proposed from different disciplines. Some papers [79, 82] have provided comprehensive introductions to these notions. Here we list some of the most widely used notions in recommender systems.

- *Equal Opportunity* [47] requires a model to produce the same true-positive rate for all individuals or groups. Equal opportunity fairness has already been recognized as unquestionably important in recommender systems by previous works. *Equalized Odds* defines the constraint that the false-positive rate and true-positive rate should be equal for the protected group and advantaged group, which represents the equal opportunity principle.
- *Envy-freeness* [29]: Every (group of) user(s) should prefer their recommendations to the recommendations of other (groups of) users. Envy-freeness guarantees neither the parity of recommendations nor equal user satisfaction.
- *Demographic Parity* [138], also called independence or statistical parity, requires that decisions should be similar around a sensitive attribute such as gender or nationality. The flaw is that demographic parity will cause a loss in utility and also infringes on individual fairness.
- *Counterfactual Fairness* [70] is an individual-level causal-based fairness notion. It requires that for any possible individual, the predicted result of the recommender system should be the same in the counterfactual world as in the real world.

- *Fairness through Awareness* [5] is also called individual fairness. It requires similar recommendations for similar individuals. In other words, any two individuals who are similar with respect to a similarity metric should receive a similar recommendation.
- *Fairness through Unawareness* [124]: Suppose we known some protected attributes, such as gender; however, if our model is unaware of these attributes, then the recommendations are regarded as fair. This approach has been shown to be ineffective in many cases, because protected variables could be correlated with other variables in the data.

In addition to the fairness notions listed above, some papers also use other fairness notions to adapt to their scenarios. For example, in Reference [91], *egocentric fairness* is proposed, which requires that the impact of the changed system should be limited or small. In Reference [150], the *Rawlsian Max-Min fairness principle of distributive justice* is applied, which requires a model to maximize the minimum utility of individuals or groups so that no subject is under-served by the model. This fairness accepts inequalities and thus does not require the decreasing utility of better-served subjects.

2.4 Fairness in the Recommender Systems vs. Fairness in Machine Learning

Machine learning is a method to uncover hidden patterns and insights from data and is a branch of artificial intelligence. Recommender systems are important applications of machine learning algorithms that offer “relevant” suggestions to users.

If we only care that the algorithms in a recommender system can precisely predict the degree that a user likes an item, then the recommendation algorithms are typical machine learning algorithms. However, the targets of recommender systems often are not limited to precision metrics. Instead, recommender systems should balance the interests of all stakeholders in the business environment. At the same time, recommender systems should balance the short-term and long-term interests of users.

Fairness is a hot research topic in machine learning. When algorithms in a recommender system are only regarded as machine learning algorithms, the fairness issues in recommender systems and machine learning are the same. Actually, some studies adopt this viewpoint. But for other studies, the interests of different stakeholders are considered so fairness in recommender systems is more complex than in machine learning.

3 FAIRNESS METRICS IN RECOMMENDER SYSTEMS

Fairness in recommender systems can be classified from different perspectives. So far, Reference [93] offers the most comprehensive classification, which classifies fairness from level (individual and group), side (consumer and producer), and graduality (single output and multiple output). We propose a more comprehensive framework to classify fairness metrics from four dimensions as follows:

- Fairness for whom
- Demographic unit
- Time frame
- Quantification method

Fairness for whom indicates which stakeholders of the system are considered. *Demographic unit* shows whether the metric considers the fairness of the group or the individual. *Time frame* indicates whether the system considers the fairness of a single-round recommendation or long-term fairness. *Quantification method* shows how the fairness metrics is defined and calculated. In the

following subsections, we introduce each dimension in detail, as well as the related work along this dimension.

3.1 Fairness for Whom

There are different participants in recommendations and some of them have fairness requirements [15]. According to the stakeholders that the system mainly considers, we can divide it into the fairness of users, providers, side stakeholders, and multiple sides.

3.1.1 Users. It is the stakeholder for which most fairness metrics are defined in recommender systems, since the interests of users are the focus of these systems. The interests of users in recommendations are mainly related to recommendation quality, which is usually reflected in the extent to which the recommendation results match the users' preferences, whether the user has adopted the recommended results or the diversity of the recommended results, and so on. Then to ensure fairness requires the recommendation quality to users is equal. However, there are too many different recommendation quality metrics, which leads to various definitions of user fairness. Another issue is that the user's perceived quality of recommender systems is not consistent with the measures of statistical quality [25]. More seriously, different users employ different criterion to judge recommendation quality, which makes the satisfaction of users almost impossible to evaluate. Therefore, currently, all the fairness metrics for users are defined based on statistical quality rather than the true satisfaction degree of users.

3.1.2 Providers. In a broader sense, providers include the authors of artwork, the sellers of products, manufacturers, and service providers. A platform should respect different providers' interests so that it can attract more providers and more items. At the same time, a platform may charge providers, which obligates it to treat providers fairly.

The rights of item providers in recommender systems are reflected in the exposure of their items in the system. They hope their items can be recommended to users more frequently and be placed as high as possible on recommendation lists, which brings more attention, benefits, and influence.

Provider fairness can be in consistent with user fairness when recommendation quality is measured in terms of the diversity of providers. But in most cases, increasing provider fairness may do harm to user fairness.

3.1.3 Side Stakeholders. The side stakeholders also experience a fairness problem caused by recommendation. Using takeout food as an example, there is a sharp increase in the number of orders for a restaurant due to recommendations, the couriers of the third-party company who are responsible for this restaurant may suffer from much higher workloads than others. This is also an indirect kind of unfairness brought by recommendations.

Currently, there is little research on the fairness problem brought by recommender systems to other stakeholders.

3.1.4 Multi-sided Fairness. While most studies only consider the fairness of one party, a small number of papers also consider the fairness of two parties, i.e., users and providers, at the same time (see Table 3). Since there are more stakeholders in recommender systems, it is possible to have different combinations of stakeholders whose fairness is considered at the same time. So far, there is no research on the fairness of other combinations except for users and providers.

3.2 Demographic Units

A demographic unit to which a recommendation is made can be divided into individual or group. Individual fairness emphasizes the similar treatment of similar individuals [95]. For example, for

any two users u_i and u_j , their treatment measurements S_{u_i} and S_{u_j} are equal as follows:

$$S_{u_i} = S_{u_j}, \forall u_i, u_j \in U. \quad (1)$$

Group fairness emphasizes that the benefits of the group or the probability of receiving services accord with the demographic structure. The construction of a group can be based on protected attributes, such as gender, race, age, and so on, or it can be a naturally formed group, such as a group in a social network.

3.3 Time Frame

In a platform, recommendations can be provided to customers repeatedly over a period of time. In some application scenarios, such as an advertising push, only the performance of a single round recommendation is concerned, so we only care about the fairness level for the single round recommendation. However, in the situation of lifelong recommendations, it is the accumulated fairness we care about. This means the lower satisfaction degree of users in a single round recommendation can be compensated for later so that they can receive accumulated fair treatment.

3.4 Quantification Method

The three dimensions define the target group and the scope in which fairness is concerned. After they are determined, fairness should be measured quantitatively. The level of fairness requires that certain interests between individuals or groups in the system are equal to each other. However, equivalence is difficult to measure directly in practical applications. Therefore, most studies measure the degree of unfairness of the system instead.

Fairness can be explicitly or implicitly measured in quantitative ways. When it is explicitly calculated, most of its calculation processes can be divided into two stages. The *first stage* is the measurement of performance metrics related to the interests of stakeholders. The *second stage* is to measure the degree of dispersion of metrics over stakeholders. Nevertheless, the second stage can be omitted, i.e., performance metrics can be used to indicate fairness directly, such as in Reference [103]. For implicit approaches, fairness is not measured directly; instead, we try to prove the factors that can cause unfairness are removed.

For ease of expression in the rest of this article, we use the following notations:

- $U = \{u_1, u_2, \dots, u_m\}$ is the user set.
- $I = \{I_1, I_2, \dots, I_n\}$ is the item set to be recommended.
- $P = \{p_1, p_2, \dots, p_c\}$ is the set of providers of items.
- I_{p_i} is the set of items provided by provider p_i .
- $R = (r_{u_i, I_j})_{m \times n}$ is a relevant rating matrix. r_{u_i, I_j} represents the ground truth and $\hat{r}_{u_i, I_j}$ is produced by the original recommendation algorithm of the system so that they compose a predicted rating matrix $\hat{R}$.
- k is the length of the recommendation lists.
- $L^g = \{l_{u_1}^g, l_{u_2}^g, \dots, l_{u_m}^g\}$ is a set of ideal recommendation lists if the users preference is known in advance or generated based on ground truth directly.
- $L^o = \{l_{u_1}^o, l_{u_2}^o, \dots, l_{u_m}^o\}$ is a set of original recommendation lists based on $\hat{R}$ and other information.
- $L = \{l_{u_1}, l_{u_2}, \dots, l_{u_m}\}$ is a set of recommendation lists finally outputted to users.
- $l_{u_i}[t]$ is the t th item in the list of u_i , $l_{u_j}^i$ is the top i item list in l_{u_j} .
- $y(u_i, I_j)$ is the click feedback for I_j from user u_i . It is 1 if clicked; otherwise, it is 0.
- $A = a_1, a_2, \dots, a_s$ is the set of disjoint protected attribute values (each candidate has exactly one value in A) either for users or providers.
- s_{u_i} represents the performance score for u_i based on some performance metrics.

3.4.1 Performance Metrics. When metrics are used to evaluate the performance of recommender systems, many of them rely on both the ground truth and the outputs from recommender systems, such as precision or recall. At the same time, some metrics can only rely on the outputs of recommender systems, such as diversity. When metrics are used for fairness evaluation, in addition to the ground truth and the original outputs of recommender systems, sometimes the same metrics can also be applied to the manipulated recommendation results against the original recommendations.

The performance metrics are the indicators that the stakeholders are concerned in the recommendation processes. Since different stakeholders have their own concerns, we discuss these separately.

User-oriented metrics. The quality of the recommendations is the main focus of users and user-oriented metrics are centered around recommendation quality.

(1) Cumulative gain. The most intuitive quality requirement for a user is that more preferred items should have a higher probability of appearing in the recommendation. Therefore, if the preference score of an item can be predicted by the recommender system, then the optimal recommendation should include the items with the top scores. The cumulative gain, which simply adds the user's preference scores for all items in the recommendation list to indicate the user's satisfaction with the recommendation results is widely applied [78, 110, 130, 149]. For a user u_i , the cumulative gain is defined as follows:

$$CG_{u_i} = \sum_{j \in I_{u_i}} \hat{r}_{u_i, I_j}. \quad (2)$$

(2) Rating prediction errors. Rating prediction accuracy is the mostly used quality metric in the recommendation system literature.

Root mean squared error is the most popular metric used in evaluating accuracy of predicted ratings while **mean absolute error (MAE)** is an alternative. In Reference [98], **mean squared error (MSE)** is applied for fairness measurement.

In addition, **underestimation error (UE)**, which measures inconsistency in how much the predictions underestimate the true ratings, and **overestimation error (OE)**, which measures inconsistency in how much the predictions overestimate the true ratings, can also be calculated [138]. The calculations of UE and OE for u_i are listed as follows:

$$UE_{u_i} = \frac{1}{|I_{u_i}|} \sum_{j \in I_{u_i}} \max(0, r_{u_i, I_j} - \hat{r}_{u_i, I_j}), \quad (3)$$

$$OE_{u_i} = \frac{1}{|I_{u_i}|} \sum_{j \in I_{u_i}} \max(0, \hat{r}_{u_i, I_j} - r_{u_i, I_j}). \quad (4)$$

(3) Interaction predictions. A user may click the link of a recommended item or buy the item directly, i.e., he or she chooses this recommended item.

Click-through rate (CTR), the ratio between the number of clicked items and the number of items that a user views, is adopted in Reference [120]. In addition, we have four possible prediction outcomes when a set of items are predicted that include true positive, false negative, false positive, and true negative.

Usage prediction performance such as *Precision@k* and *Recall@k* can be measured by counting the number of examples that fall into each of the above four categories.

More complex measurements, including **receiver operating characteristic (ROC)** curves, **area under the ROC curve (AUC)**, and **F1-score** [127], can also be applied.

(4) Ranking measures. It is hoped that in the recommendation list, the more relevant items should be ranked higher [134]. There are many metrics for ranking quality.

Pairwise accuracy measures the probability that a clicked item is ranked above another relevant unclicked item for the same query [6, 119],

$$P(c(I_i, I_j, u_t) | y(u_t, I_i) > y(u_t, I_j)), I_i, I_j \in l_{u_t}, \quad (5)$$

where $c(I_i, I_j, u_t)$ represents item I_i is ranked above item I_j for user u_t .

Kendall's τ is used to measure the correctness of the relative ranking of pairs of items [103]. A pair of items is concordant if their relative ranking is the same in both lists and discordant if their relative ranking is reversed in the lists. *Kendall's τ* is calculated as follows:

$$\tau = \frac{C - D}{C + D}, \quad (6)$$

where C is the number of concordant pairs and D is the number of discordant pairs.

The **average precision (AP)** of a list is the average of precision at every recall level [103],

$$AP = \frac{1}{I_g} \sum_t Precision@t \cdot |l_{u_i}[t] \cap L_{u_i}^g|. \quad (7)$$

Normalized discounted cumulative gain (NDCG) is a popular metrics to measure the utility of the whole list, where positions are discounted logarithmically. **Discounted cumulative gain (DCG)** for top- k list is calculated as follows:

$$DCG_{u_i} = \sum_{j=1}^k \frac{2^{r_{u_i, l_{u_i}[j]}} - 1}{\log_2(j + 1)}, \quad (8)$$

$$NDCG_{u_i} = \frac{DCG_{u_i}}{IDCG_{u_i}}. \quad (9)$$

In Reference [103], *Kendall's τ* , *AP*, and *NDCG* are all applied to measure fairness.

(5) Distribution conformity. In the historical information of users, the distributions of classes of items with some attribute values reflect their preferences. In other cases, the distributions of classes correspond to the anticipated quality of recommendations. Therefore, the difference between the actual distribution of item classes and the desired one can be used to measure recommendation quality.

The items in the recommendation list is a discrete probability distribution over the set of classes. To compute the extent to which the set of top k ranked results for a recommendation task differ over an attribute value with respect to the desired proportion of that attribute value, the *relative ratio in recommendations* can be computed. Moreover, *Skew _{a_i}* is the logarithmic relative ratio [42],

$$Skew_{a_i}(l_{u_j}) = \log_e \frac{p(l_{u_j}, a_i)}{p(l, a_i)}, \quad (10)$$

where a_i is an attribute value, $p(l, a_i)$ is the proportion of items with attribute value a_i , and $p(l_{u_j}, a_i)$ is the proportion of items with attribute value a_i in the u_j 's recommendation list l_{u_j} .

For two distributions, we can use **Kullback–Leibler (KL)** divergence to represent their difference,

$$d_{KL}(D_1 || D_2) = \sum_j D_1(j) \log_e \left(\frac{D_1(j)}{D_2(j)} \right), \quad (11)$$

where $d_{KL}(D_1 || D_2)$ is the *KL divergence* of distribution D_1 with respect to distribution D_2 .

In Reference [123], *KL divergence* is used to measure fairness in location recommendation.

A **normalized KL divergence (NKL)** is defined in Reference [104],

$$D(l_{u_j}) = 1 - \frac{d_{KL}(D_{l_{u_j}} || D_1)}{d_{KL}^+(\cdot || D_1)}, \quad (12)$$

where $d_{KL}^+(\cdot || D_1)$ is the upper bound on the divergence of any possible representation.

Moreover, in Reference [42], **normalized discounted cumulative KL divergence (NDKL)** is proposed to measure distribution conformity,

$$NDKL(l_{u_j}) = \frac{1}{Z} \sum_{i=1}^{|l_{u_j}|} \frac{1}{\log_2(i+1)} d_{KL}(D_{l_{u_j}}^a || D_1), \quad (13)$$

where $D_{l_{u_j}}^a$ is the discrete distribution assigned to each attribute value in A (the proportion of candidates with that value) over the top i candidates in the given ranked list l_{u_j} , D_1 is the desired distribution, and $Z = \sum_{i=1}^{|l_{u_j}|} \frac{1}{\log_2(i+1)}$.

(6) Discrepancy between recommendations. The actual recommendations can be different from the ideal ones for several some reasons. Therefore, it is possible to compare the discrepancy of metrics between manipulated and original recommendations.

As an example, the ratio of cumulative gain in the recommendation list to the cumulative gain in the original recommendation list can be calculated, denoted as *score similarity* [63, 110],

$$D_{u_i}^C = \frac{CG_{u_i}}{CG_{u_i}^o}. \quad (14)$$

In Reference [63], a simpler discrepancy measurement is defined, i.e., the similarity among the recommended items to users and their top- k items with respect to preference scores, denoted as *recommendation similarity*,

$$D_{u_i}^R = \frac{l_{u_i} \cap l_{u_i}^o}{k}. \quad (15)$$

(7) Others. There are many other metrics defined for recommender system evaluation, such as *diversity*, *serendipity*, and *novelty*. *Diversity* is adopted to measure fairness in Reference [56]. The other metrics have not been applied.

Provider-oriented metrics. For providers, what they care about is that the items they provide have more exposure and visibility. Since true exposure or visibility is hard to measure, we can use the positions of their items in the list as an index. A more straightforward way is to count the total number of times that their items appear in the recommendation list.

(1) Exposure-related metrics. This category of metrics takes the appearance chance or number of times of the items appear in the recommendation lists into consideration.

Recommended times counts how many times the items from a provider are recommended directly.

Rank-Weighted Exposure [7] weights the exposure of an item by its rank in the list,

$$e_i = \frac{1}{\log_2(r_i^l + 1)}, \quad (16)$$

where r_i^l is the rank of item i in list l .

Coverage is used to measure the percentage of all providers whose items are recommended. In [76], **average provider coverage rate (APCR)** is defined as follows:

$$APCR = \frac{\sum_i^m P(l_{u_i})}{cm}, \quad (17)$$

where $P(l_{u_i})$ is the number of providers in list l_{u_i} .

In Reference [11], the *outcome of item* is defined as the probability that an item is ranked in the recommendation list, and the *true-positive rate* is the probability of being ranked within the top- k of a user, given the ground truth that the item is relevant for that user in the test set.

Pairwise accuracy is also related to exposure, since it measures whether items from a group (a provider) are under ranked systematically.

(2) Disparity between exposure and relevance. Emphasizing absolute fairness in providers' exposure neglects the requirements of users. Therefore, a more balanced approach is to ensure the exposure of items is proportional to their relevance in a series of rankings.

The *exposure-based fairness disparity* expresses the average proportion of exposure got by each group to its relevance over all timesteps [85].

Similarly, *impact-based fairness disparity* expresses the average proportion of feedback (e.g., clicks, purchases, votes) obtained by each group to its relevance over all timesteps [85].

The *abstract distance between attention and relevance* can be calculated as follows [7, 13]:

$$D_{u_i, I_j}^{tr} = \left| \sum_{j=1}^n e_j - \sum_{j=1}^n r_j \right|, \quad (18)$$

where e_j is the attention of item I_j received from users, which is the exposure of item I_j .

(3) Distribution conformity. Distribution conformity metrics introduced above can also be applied to provider fairness. Items can be divided into categories in terms of providers or attribute value. Ideally, the appearance probability distribution of items of different categories should be consistent to their percentages shown in the platform.

(4) Diversity. Diversity can measure provider fairness in a rough manner.

The providers can be divided into different groups based on their position in the popularity spectrum. They can be divided into bins of equal size in order of popularity. Then the *diversity of providers with different popularity* can be calculated as follows [83]:

$$\psi(l) = \sum_{i=1}^h \sqrt{|P_i \cap P(l)|}, \quad (19)$$

where h is the number of bins, P_i is all the providers belonging to i th bin, and $p(l)$ is all the providers in the list.

Popularity rate refers to the proportion of popular items in the recommendation list against the total number of items in the list, which can be seen as a popularity-level measure of fairness [39].

3.4.2 Synthesize Measurement Approaches. After the performance metrics are measured, some of them can be used as the final fairness measurement, but more of them should be synthesized over all populations.

The definition of fairness is basically to hope that certain properties can remain the same or as close as possible between entities. Generally, the fairness level is difficult to compute while unfairness is easier to calculate. Therefore, the measurement of fairness in related research is based on the degree of dispersion between data. The greater the degree of dispersion of the data, the more unfair the results and vice versa.

There are many traditional statistical metrics that can be used to calculate the fairness or unfairness over populations, such as *mean*, *sum*, *variance*, *entropy*, *mean absolute difference*, *absolute mean difference*, *L1 norm*, *L2 norm*, *Mean Squared Difference*, and so on. In addition, some specific metrics are also used.

(1) *Gini coefficient*, also known as the *Gini index* or *Gini ratio*, is a measure of the demographic distribution with the aim of projecting the income of a nation's populace. The *Gini coefficient* has been the most commonly used estimation of inequality, and it is also used widely in fairness measurement. The inverse of the *Gini index*, which is one minus the *Gini index* value, indicates the equality is also applied.

(2) *Jain's Fairness Index* [146] measures the appearance differences of a set of values as follows:

$$J = \frac{(\sum_{i=1}^n x_i)^2}{n \sum_{i=1}^n x_i^2}, \quad (20)$$

where x_i is the number of times a provider's items appear (or a class of items, items with some attribute values).

(3) *Mean average envy* [91]. Envy-freeness is a classical fairness notion. The mean average envy is defined as follows:

$$E = \frac{1}{n} \sum_{u \in U} \frac{1}{n-1} \sum_{u' \in U, u' \neq u} \text{envy}(u, u'), \quad (21)$$

where $\text{envy}(u, u')$ is the extra utility u would have received if she had given the result that had been given to u' instead of her own allocated result.

(4) *Disagreement of the group* [110]. This approach can measure unfairness in a group. It is the difference in the overall satisfaction scores between the most-satisfied and the least-satisfied member in the group.

(5) *Discrepancy between two Groups*. For any two groups, we can calculate their discrepancy as the unfairness measurement. For example, we can calculate the *difference between means of different groups*. Moreover, we can use metrics to calculate the discrepancy between two distributions, for example, the traditional *KL divergence*. In Reference [149], another metrics for the equality of two distributions, the **Kolmogorov-Smirnov statistic (KS)** is applied.

3.5 Implicit Approach

Sometimes, fairness is not measured directly. For example, in Reference [70], fairness is evaluated in terms of whether an attacker can predict the sensitive feature value based on the filtered user embeddings. In Reference [124], the prediction performance of the sensitive user attribute based on the top K ranked candidate news in each session is regarded as the indication of recommendation fairness. If using a sensitive user attribute does not increase performance, then it is fair.

3.6 Instruction of Proposed Framework and Statistics of Applications of Fairness Metrics

In this section, we describe how readers can use our proposed framework to find fairness metrics that meet their needs. The instruction is show in Figure 2. The method is also very simple, that is, to determine the details of the four dimensions of the framework. For a reader who wants to evaluate the fairness of a recommender system, he or she first needs to determine which side of the stakeholder fairness he cares about, that is the *fairness of whom* dimension. For example, a shopping recommendation platform may pay more attention to the fairness of the recommended items, that is, the provider-side fairness, and a work recommendation platform may pay more attention to the user-side fairness. Furthermore, both users and providers are essentially the customers of the

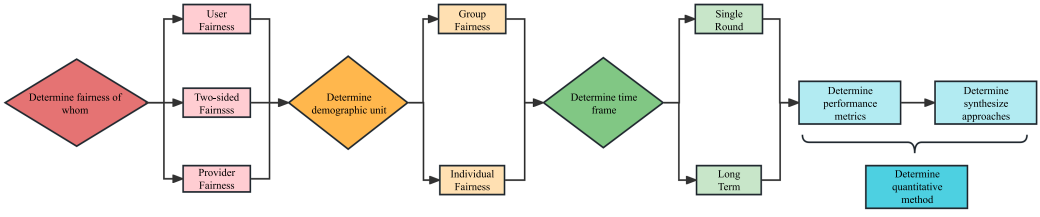


Fig. 2. Instruction of proposed framework.

platforms, and they are both important stakeholders in the recommender systems, so more and more studies choose to evaluate the fairness of both stakeholders at the same time.

The next step is to determine the *demographic unit*. It depends on whether you need to consider differences in interests between individuals or between groups. Generally speaking, under the premise of considering the fairness of users, it is more intuitive to consider group fairness, because the basic attributes of users (such as race, nationality, gender, etc.) naturally divide users into different groups. The unfairness of interests between these groups divided by basic attributes is also the issue that first triggered the discussion of fairness in recommender systems. Then, for the case of choosing provider fairness, it may be more straightforward to consider individual fairness, because the recommended items are more like independent individuals being compared with each other in the recommender system. Of course, it is also reasonable to consider the individual fairness from the user side and the group fairness from the provider side, which needs to be selected according to your own needs.

The choice of *time frame* is relatively independent of the first two dimensions and depends more on the type of recommender system. For example, an advertising recommendation system may pay more attention to the fairness of a single recommendation, and an online shopping recommendation system may pay more attention to the long-term fairness.

The final step is to determine *quantification method*. This part can be divided into two stages, namely determining performance metrics and determining measurement approaches. Performance metrics generally refer to direct metrics produced by recommender systems, which we categorize and elaborate on in Section 3.4.1 according to the stakeholders considered. The fairness metrics is a higher-level indicator based on performance metrics, which is usually an aggregation of performance metrics or the degree of dispersion between data. Therefore, after determining the performance metrics to be considered, it is often only necessary to further determine the method of aggregation or the method of evaluating the degree of dispersion of the data, that is, the synthesize measurement approaches in Section 3.4.2, to determine the quantitative method of fairness.

Next, we further strengthen readers' understanding of our framework by showing how our framework fits into existing research on fairness in recommender systems. At the same time, through the classification of existing research, readers can easily find fairness metrics that suit their needs. We also draw some statistical conclusions based on the collation of related studies. We list the fairness metrics for recommendations in the literature in Tables 1–3. Although the lists are not inclusive, they represent the overall status, since they cover most of the important work in recent years.

The amount of research on group fairness and individual fairness is almost the same, which means in practice, group fairness and individual fairness are both important topics. For the time frame, it can be seen that, for user fairness, the studies mainly focus on single-round recommendation. In contrast, for provider fairness, the numbers of studies on long-term and single-round fairness are almost equal. The reason for this may be that the market requires that fairness should be ensured for each round. However, providers care more about long-term fairness.

Table 1. User Side

Reference	year	Demographic Unit	Time Frame	Fairness Notion	Performance Metrics	Synthesize Measurement
[138]	2017	Group	Single Round	Equal Opportunity	Mean Errors, MAE, Overestimation Errors, Underestimation Errors, Overall Predictions	Absolute Mean Difference
[63]	2018	Individual	Single Round	Fairness through awareness	Score Similarity, Recommendation Similarity	Gini Coefficient
[98]	2019	Individual and Group	Single Round	Fairness through awareness and Demographic parity	MSE	Variance
[123]	2019	Group	Single Round	Equal Opportunity	KL Divergence	KL Divergence
[78]	2019	Individual	Single Round	Fairness through awareness	Cumulative Gains	Mean Absolute Deviation
[104]	2019	Individual	Single Round	Fairness through awareness	NKL	Mean
[103]	2019	Group	Single Round	Rawlsian Max-Min fairness	Kendall's τ , AP, NDCG	—
[110]	2020	Individual	Long Term	Fairness through awareness	Cumulative Gains	Disagreement of the Group
[130]	2020	Individual	Long Term	Fairness through awareness	Cumulative Gains	Variance
[120]	2020	Group	Single Round	Fairness through awareness	CTR	Gini Coefficient
[36]	2020	Individual and Group	Single Round	Fairness through awareness and Demographic parity	NDCG	Absolute Mean Difference
[69]	2021	Group	Single Round	Demographic parity	Preference score	Absolute Mean Difference
[124]	2021	Group	Single Round	Demographic parity	Accuracy and F1-score	—
[53]	2021	Group	Single Round	Demographic parity	Preference score	Absolute Mean Difference
[127]	2021	Group	Single Round	Demographic parity	AUC and F1-score	—
[29]	2022	Individual	Single Round	Envy-freeness	Cumulative Gains	Mean average envy
[106]	2022	Individual	Single Round	Envy-freeness	Cumulative Gains	Average
[122]	2022	Individual and Group	Single Round	Fairness through awareness, Counterfactual Fairness and Demographic parity	MAE, NDCG and AUC	Discrepancy between two Groups
[96]	2022	Group	Single Round	Demographic parity	NDCG	Discrepancy between two Groups
[114]	2023	Group	Single Round	Demographic parity	Preference score	Discrepancy between two Groups
[71]	2023	Group	Single Round	Counterfactual Fairness	AUC	—

The most common fairness notion applied is fairness through awareness, which fits for individual-level fairness. For group-level fairness, different fairness notions are applied.

The performance metrics adopted in the research are various, which reflects different angles for the evaluations of recommender systems. It is partly attributed to different contexts in which recommendations are applied.

3.7 Desired Properties of Fairness Metrics

Obviously, the metrics for fairness are versatile. Instead of seeking a definition of a unified fairness metric set, we can try to define some rules to evaluate the appropriateness of metrics.

In Reference [109], the axioms borrowed from economics and social science for measuring inequality are overviewed, and they can be applied to evaluate the fairness metrics.

Anonymity: The inequality measure should not depend on the characteristics of attributes except for their values obtained by the model.

Zero normalization: The inequality measure should be minimized when all attribute values are equal (i.e., the uniform distribution). The minimum value for the fairness metric should be zero.

Population invariance: The measure is independent of the size of the population under consideration.

Transfer principle: Transferring the benefit from a high-benefit attribute value to a low-benefit value, if it does not reverse the relative position of values, must decrease the inequality.

Table 2. Provider Side

Reference	year	Demographic Unit	Time Frame	Fairness Notion	Performance Metrics	Synthesize Measurement
[7]	2018	Individual	Long Term	Fairness through awareness	Abstract Distance between Attention and Relevance	Summation
[16]	2018	Group	Single Round	Equal Opportunity	Recommendation Times	Ratio Difference
[56]	2018	Group	Single Round	Demographic parity	Relative Ratio in Recommendations	–
[146]	2018	Individual	Long Term	Fairness through awareness	Recommended Times	Jain's fairness index
[83]	2018	Group	Single Round	Demographic Parity	Diversity of Providers with Different Popularity	–
[13]	2019	Individual	Long Term	Fairness through awareness	Abstract Distance between Attention and Relevance	Mean
[6]	2019	Group	Single Round	Equal Opportunity	Pairwise Accuracy	–
[42]	2019	Group	Single Round	Equal Opportunity and Demographic Parity	Skew, NDKL	Mean
[35]	2019	Individual	Long Term	Fairness through awareness	Coverage, Recommended Times	Gini Coefficient
[76]	2019	Group	Single Round	Demographic Parity	APCR	–
[85]	2020	Group	Long Term	Demographic Parity	Exposure-based Fairness, Impact-based Fairness	Mean
[12]	2021	Group	Single Round	Demographic parity	Coverage	–
[11]	2021	Individual	Single Round	Fairness through awareness	Outcome of Item, True Positive Rate	Inverse of Gini Coefficient
[38]	2021	Individual and Group	Single Round	Rawlsian Max-Min fairness	DCG	Gini Coefficient
[121]	2021	Group	Single Round	Counterfactual Fairness	Outcome of item	–
[148]	2021	Individual	Long Term	Fairness through awareness	True Positive Rate	Gini Coefficient
[150]	2021	Individual	Single Round	Equal opportunity and Rawlsian Max-Min fairness	True Positive Rate	Mean
[46]	2021	Group	Single Round	Demographic Parity	Rank-Weighted Exposure	Discrepancy between two Groups
[119]	2021	Group	Single Round	Demographic Parity	Rank-Weighted Exposure	Discrepancy between two Groups
[39]	2021	Individual	Single Round	Fairness through awareness	Recommended Times, Popularity Rate	Gini Coefficient
[40]	2022	Group	Single Round	Demographic Parity	NDCG	KL divergence
[94]	2022	Group	Single Round	Demographic Parity	Recommended Times	Ratio Difference
[68]	2022	Individual	Single Round	Fairness through awareness	Rank-Weighted Exposure	Mean Absolute Difference
[41]	2022	Group	Single Round	Demographic Parity	Recommended Times	KL divergence
[126]	2022	Individual and Group	Single Round	Demographic Parity	NDCG (Customer), Outcome of item(Provider)	Discrepancy between two Groups

Additive property: For long-term recommendation, where multiple rounds of recommendations are made, the overall fairness over the process is defined in terms of the addition of the fairness metrics of each round. Similarly, if the fairness metrics for a group can be calculated in terms of the addition of the fairness metrics of individuals, then we call it the additive property over populations.

Equivalence of Fairness across Different Domains: Since the platform regards recommendation as a service, recommendations are often provided in multiple domains. Although it is difficult to ensure fairness in all domains, we can ensure overall fairness over all domains. Thus, there is a requirement that the fairness scores in different domains can be exchanged in some form.

4 FAIRNESS ASSURANCE STRATEGIES IN RECOMMENDER SYSTEMS

In this section, we introduce in detail the algorithm for eliminating unfairness in recommendation. Roughly, the algorithms can be divided into pre-processing, in-processing, and post-processing according to the stages of the algorithm's operation in the algorithm process.

Table 3. Two-side Fairness

Reference	year	Demographic Unit	Time Frame	Fairness Notion	Performance Metrics	Synthesize Measurement
[149]	2018	Group	Single Round	Demographic parity	Cumulative Gains	Absolute Mean Difference, KS
[112]	2019	Individual	Long Term	Fairness through awareness	Utility (Waiting Time (Customer), Income(Driver))	Entropy
[90]	2020	Individual	Long Term	Fairness through awareness	Cumulative Gains (Customer), Exposure(Provider)	Entropy of Producer Exposures, Mean Average Envy, Standard Deviation of Customer Utilities
[91]	2020	Individual and Group	Single Round	Egocentric Fairness	Score Similarity (Customer), Exposure (Provider)	L1 Norm
[117]	2020	Group	Single Round	Equal Opportunity	Prediction Errors (Customer), Distribution Conformity(Provider)	Variance (Customer)
[131]	2021	Individual and Group	Long Term	Fairness through awareness and Demographic Parity	NDCG (Customer), Exposure (Provider), Exposure-based Fairness Disparity (Provider)	Variance
[86]	2022	Group	Single Round	Demographic Parity	NDCG (Customer), Exposure (Provider)	Discrepancy between two Groups

The pre-processing algorithm is applied to eliminate bias in the data. The in-processing algorithm usually improves the traditional recommendation algorithm to obtain fairer results. The post-processing algorithm improves the fairness of the recommendation results obtained by the existing recommendation algorithm.

4.1 Pre-processing

The pre-processing strategies focus on the data itself. It tries to remove the bias from the data so that the recommendation algorithm can produce fairer results.

Pre-processing techniques can roughly be divided into four categories [55], i.e., *suppression*, *massaging* the dataset, *reweighing*, and *sampling*. Of these, reweighing means the data records in the training dataset are assigned weights, and we can adjust the weight to remove the bias if the algorithm is based on frequency counts. Since reweighing does not change the data themselves, we put them into the in-processing category.

- **Suppression:** If the recommender system uses the attributes that correlate most with the sensitive attribute, then we can remove the sensitive attribute and those most correlated attributes. However, when some attributes are removed, it may greatly lower the recommendation quality.
- **Massaging the dataset:** In this approach, the labels of some data records are changed. Obviously, it is rather intrusive as it changes the labels of the data records. For example, the interaction records of some users are changed. Although this may help improve the fairness for other users, the users whose records are changed may suffer a loss.
- **Sampling:** To remove bias, we can under-sample the dominant ones or over-sample the inferior ones.

In Reference [84], a data resampling strategy is applied to resample the users of the minority group (female) in the training set until they match the number of training data points of the majority group (male). When a user is resampled, his or her data are duplicated and used fully in the training procedure. The empirical study shows that although fairness is improved, an overall decrease in performance is also observed. Moreover, of the different recommendation algorithms, the one that has the better performance has the worse fairness results. This confirms the fact that the

general performance metrics of recommendation inherently conflict with fairness metrics. In Reference [34], similar findings are obtained, i.e., resampling so that each group has the same number of ratings that may increase fairness but reduce the recommendation quality.

In practice, the resampling strategy is often applied with other techniques. For example, in References [11, 12], resampling is used with regularized optimization.

The resampling strategy can be applied to a case where groups are defined according to one attribute or a few attributes, so that it is possible to balance the number of groups. When multiple attributes are related to group definitions, it is not possible to balance the data through resampling due to their internal relationships. For example, in Reference [34], it is found that correcting for one effect (popularity bias) can change the results for another effect (demographic bias).

As massaging the dataset may introduce certain problems, Reference [98] proposes an approach to augment the original data by generating some artificial users (or items) to improve the social desirability of recommender systems. These artificial data are called “antidote” data. The algorithm starts from initial antidote data with the size of a given budget. At each iteration, the factorization algorithm is applied jointly on the original data and the current antidote data to find updated factors and estimated ratings. Then the gradient of the antidote utility with respect to antidote data at the current point is computed, and the algorithm chooses a step size and updates the antidote data. After each update, a projection function is applied to obtain a feasible solution.

Pre-processing can be combined with other techniques, for example, in Reference [12], a treatment is devised that combines observation upsampling and loss regularization. Currently, there is scant research solely on pre-processing methods, which may be due to the high risk of processing experimental data.

4.2 In-processing

The in-processing strategies try to generate fairer recommendations through the recommendation algorithm redesigns. The strategies can be divided into reweighting the neighborhoods, debiasing the latent factors or embeddings, adding regularization terms relating to unfairness and multiple objectives optimization.

4.2.1 Reweighting the Neighborhoods. One of the most successful approaches to build recommender systems, **collaborative filtering (CF)** [111] uses the known preferences of a group of users to make recommendations or predictions of the unknown preferences of other users.

One of the fundamental ideas of collaborative recommendation is that of the peer user, a neighbor whose patterns of interest match those of the target user and whose ratings can be extrapolated to make recommendations for the target user. Similarly, peer items can also be used to make recommendations. One place where bias may creep into collaborative recommendation may be through the formation of peer neighborhoods.

In Reference [16], an extension of the well-known **Sparse Linear Method (SLIM)** is explored. SLIM is well known as a state-of-the-art technology for collaborative recommendation. It is a generalization of item-based recommendation in which a regression co-efficient is learned for each user-item pair. The extension of SLIM uses regularization to control the way different neighbors are weighted (neighborhood balance term), with the goal of achieving balance between protected and non-protected neighbors for each user (or items for provider fairness).

Because the neighborhood-based approach became less popular after the latent-factor based model appeared, there are a few approaches that fall into this category.

4.2.2 Debiasing Latent Factors or Embeddings. In most recommendation algorithms, the items and users are generally mapped into a latent space or an embedding space first. The bias can go

into the latent factors and embeddings. Therefore, a straightforward idea is to remove bias from them before the consequent steps.

In Reference [53], a Neural Fair Collaborative Filtering framework is proposed to mitigate gender bias in recommending career-related sensitive items. In the first step, user and item embeddings are obtained through a NCF that is trained to predict the users' interactions with nonsensitive items via back-propagation. In the second step, CF embeddings for users from each protected group are used to find the proper bias direction, and then each user embedding is debiased by subtracting its component in the direction of the bias vector. Specifically, the user embedding p_u is debiased by subtracting its component in the direction of the bias vector,

$$p'_u = p_u - (p_u \cdot v_B)v_B,$$

where v_B is the bias vector.

Similarly, Reference [149] designs a **fairness-aware tensor-based recommendation (FATR)** framework to remove the impact of sensitive attributes. The latent factor matrices of the tensor completion model contain latent information related to the sensitive attributes, which introduces unfairness. Therefore, FATR first isolates the impact of the sensitive attribute by plugging the sensitive features into the latent factor matrix. In this way, the sensitive attributes can be explicitly represented and isolated from non-sensitive attributes in the latent factor matrix. However, there may still be sensitive information that resides in non-sensitive dimensions. To extract the remaining sensitive information that resides in non-sensitive dimensions, an additional constraint that the non-sensitive dimensions should be orthogonal to the sensitive dimensions in the sensitive latent factor matrix is added into the objective function.

Adversarial learning [77] is a technique that can be used to learn bias-free deep representations from biased data. It enforces the deep representations to be maximally informative for predicting the labels of the main task and to be minimally discriminative for predicting sensitive attributes. Thus, adversarial learning has the potential to learn bias-free user embeddings by removing the biased information about sensitive user attributes.

Reference [70] introduces a model to generate feature independent user embeddings through adversary learning. The main idea is to train a predictor and an adversarial classifier simultaneously, where the predictor aims at learning informative representations for the recommendation task, while the adversarial classifier aims at minimizing the predictor's ability to predict the protected features from the learned representation, thus the information on sensitive features is removed from the representations to mitigate discrimination. Following the adversary learning setup, they develop an adversary network that consists of two modules: a filter module that aims at filtering out the information about sensitive features from user embeddings and a discriminator module that aims to predict the sensitive features from the learned user embeddings. Reference [124] proposes a decomposed adversarial learning method with orthogonality regularization to learn bias-free user embeddings for fairness-aware news ranking. In this method, the user interest model is decomposed into two components, i.e., a bias-aware one learning user embedding that captures the biased information on sensitive user attributes, and a bias-free one that only encodes the attribute-independent information of the user's interest into bias-free user embeddings. It applies a sensitive attribute prediction task to bias-aware user embedding. Because bias-free user embedding may still contain some information related to sensitive user attribute, an orthogonality regularization method is used to further purify the bias-free user embedding.

Reference [127] proposes a Fair Graph-based RecOmmendation model for ensuring fairness of any recommendation models. Given the original embeddings from any recommendation models, the model learn a composition of filters that transform each user's and each item's original embeddings into a filtered embedding space based on the sensitive feature set. For each user, this

transformation is achieved under the adversarial learning of a user-centric graph, to obfuscate each sensitive feature between both the filtered user embedding and the sub graph structures of this user.

Based on the theoretical analysis, contrastive learning paradigm has a debiasing effect that can benefit recommender systems, and there is a connection between contrastive learning and inverse propensity weighting, where the latter is a well-studied technique for bias reduction. Reference [144] introduces *CLRec*, a contrastive learning framework for debiased deep candidate generation in recommender systems. The implementation maintains a fixed-size first-in first-out queue to accumulate positive samples and their representations from the previous batches and use the content of the queue to serve as the negative samples to be discriminated from the positive samples of the next batch. It guarantees that all items will be sampled sometime in an epoch to serve as the negative examples and allows them to reuse the computed results from the previous batches.

4.2.3 Regularization Term for Unfairness. Modifying the objective function by adding a fairness regularization term is an intuitive way to optimize the algorithm in a fairer direction during the optimization process. In related work, different regularization terms for unfairness are defined based on different fairness notions and measurement metrics.

The regularization term can be designed in terms of the unfairness measurement. In Reference [12], a regularization term is designed with the aim of penalizing the model when the difference in relevance and contribution for the minority group of providers is high. In Reference [53], the regularization term is the average of differential fairness measures for sensitive items, which addresses the demographic bias in the sensitive items.

However, regularization terms can also be designed to penalize the correlation between the predictions of items and their sensitive properties. The work in Reference [117] regularizes the correlation between prediction errors and the distribution of market segments. In Reference [6], the pairwise regularizer calculates the correlation between the residual between clicked and unclicked items and the group membership of the clicked item. As a result, the model is penalized if its ability to predict which item was clicked is better for one group than the other. In Reference [11], the regularization term is the biased correlation between the predicted residuals and the observed-item popularity to mitigate a popularity bias. The model is thus penalized if its ability to predict a higher relevance for the item depends on popularity.

4.2.4 Multiple Objective Optimization. Different from above approaches that try to improve fairness in recommendation without changing the main part of the algorithms, in some recommender systems, the main body of recommendation algorithms are changed.

Since we need to consider user satisfaction and fairness at the same time, the problem becomes a multiple objective optimization problem. When multiple objectives are concerned, a Pareto efficient recommendation is expected [102].

Scalarization is a common approach to solve the multiple objective optimization problem. In Reference [83], the tradeoff between the relevance of recommendations to the consumer (i.e., user) and fairness of representation of suppliers (i.e., artists) in the case of music streaming services is considered. The first approach represents the objective as a weighted combination of its relevance and fairness estimates. Other approaches are also suggested in this article. *probPolicy* decides whether to recommend content based on fairness or on relevance for a user according to a probability threshold. *Guaranteed policy* models the certain minimum amount of relevance as the constraint. Finally, *adaptive policy* computes a user's affinity to different types of content (especially fair sets) and adaptively leverages such user-level affinity to recommend content.

In Reference [9], a strategy is proposed to adjust the recommendation results to balance accuracy and fairness. It uses the PMF method to obtain the user and item latent factors. Then a **multilayer**

neural network (MLN) is trained with the loss function to minimize the weighted combination of the prediction error and the fairness error. The MLN returns a real number whose meaning is the predicted loss error for item i to user u recommendation. The lower the predicted loss error, the better the combined (accuracy, fairness) values.

However, the scalarization approach relies on manually assigned weights, whose Pareto efficiency cannot be guaranteed. Recently, KKT conditions were used to guide scalarization techniques. However, the existing algorithms based on these conditions are limited to unconstrained cases and cannot fit the requirements in real-world scenarios [73]. The framework in Reference [73] starts with uniform scalarization weights and then updates the model parameters and the scalarization weights alternatively. The core part of the framework is the *PECsolver*, which generates scalarization weights by solving the condition, which is a complex *quadratic programming* problem. The framework does not rely on specific formulations of the loss functions or the model structures. Although the algorithm runs with stochastic gradient descent in batches, the algorithm provides a theoretical guarantee of convergence as gradient descent.

4.2.5 Miscellaneous. Since the fairness issue in recommendation is closely related to applications, some special strategies are also designed for them.

In Reference [78], to provide fair team recommendations for all projects, in each round one applicant with the highest skill matching score is allocated to each project.

In Reference [134], after services are selected for a group through a model called the multi-facet probabilistic graph model, the positions of these services are decided based on an activeness-based coalition ranking strategy to ensure the fairness between users in the group.

In Reference [39], reinforcement learning is applied to ensure long-term fairness. The sequential interactions between consumers and recommender systems are modeled as a *Markov decision process*, and then it is turned into a *constrained Markov decision process* by constraining the fairness of item exposure at each iteration dynamically. *Constrained policy optimization* with adapted neural network architecture is leveraged to automatically learn the optimal policy under different fairness constraints.

Reference [121] proposes a model-agnostic framework MACR that performs multi-task training according to the causal graph to assess the contribution of different causes on the ranking score. A counterfactual inference is performed to estimate the direct effect from item properties to the ranking score, which is removed to eliminate the popularity bias.

4.3 Post-processing

Post-processing strategies change the original recommendation results to improve fairness levels. There are usually two solutions for post-processing algorithms. The first is to generate a new list from the candidate items one by one according to their relevance and impact on fairness. The second is to formulate specific rules to re-rank the whole list.

4.3.1 Generative Strategies. Generative algorithms usually adopt a greedy strategy by calculating the score of the candidate items and then selecting the item with the highest score to add to the list. These scores usually take both users' satisfaction and fairness into consideration. For example, Reference [56] proposes a heuristic algorithm that selects the item with the largest weighted combination of relevance and fairness from the item set each time.

Reference [57] designs a greedy algorithm to ensure fairness that "balances" the relevance of the recommended items across the group members in a rank-sensitive way. At each iteration, it inserts into the ordered result set the item from the remaining candidates that gives the highest fairness gain.

In addition to calculating the utility score of each item explicitly, the utility of each item can also be decided in an implicit way. In Reference [42], algorithms are proposed to select the item one by one in terms of which protected attribute value is going to violate the ratio constraint of the items with the specific attribute value in the whole population dynamically.

Sometimes, the post-processing cannot be completed in a single step. Multi-step generative algorithms divide the generation process into several stages, in which intermediate results are first generated. Reference [90] provides a greedy algorithm *FairRec* to achieve the desired two-sided fairness. *FairRec* includes two steps. The first step ensures envy freeness up to one item (EF1) among all customers and tries to provide a minimum guarantee on the exposure of the producers. However, the first phase may not allocate exactly k products to all users, which is then ensured by the second phase while simultaneously maintaining EF1 for users.

4.3.2 Re-ranking Strategies. In addition to the generative model, there are also approaches that directly re-rank the original recommendation lists according to certain rules. These strategies can be further divided into *heuristic re-ranking* strategies and **integer linear programming- (ILP)** based strategies.

(1) Heuristic Re-ranking Strategies. Heuristic re-ranking strategies adjust the recommendation list as a whole.

In Reference [76], the new list is built iteratively according to the following criterion: $P(i|u) + \lambda P(i, \bar{l}_u|u)$, where $P(i|u)$ is the likelihood of user $u \in U$ being interested in item $i \in I$, predicted by the base recommender. The second term $P(i, \bar{l}_u|u)$ denotes the likelihood of user u being interested in an item i not in the currently generated list l_u . The items with the highest scores are added to the output list l_u and the process is repeated until l_u has achieved the desired length.

Reference [32] designs a heuristic strategy. It tries to change the original recommendations as little as possible so the predictor has a smaller chance of predicting the sensitive attributes in terms of the recommendations at the same time. The **balanced error rate (BER)** is the average class-conditioned error of the predictor. The strategy is to select a user whose recommendation change will bring the maximal increase to BER. The approach to change the recommendation is to replace the original recommendation with a recommendation to another user whose attribute value is different.

To ensure the fairness of recommendations among members in a group, in Reference [110], a novel aggregation method that is a weighted combination of the average and the least misery aggregation methods is proposed. The post-processing consists of multiple iterations. Each iteration consists of two steps. The first step is to calculate the weight, where the recommendation list for a group is generated according to the new aggregation method, and the second step is to optimize the weight dynamically by setting the value by subtracting the minimum satisfaction score of the group members in the previous iteration from the maximum score.

Reference [148] proposes a model-agnostic False Positive Correction method for debiasing that corrects the predicted scores based on false-positive signals in a probabilistic way.

Heuristic re-ranking strategies are popular, because they are easy to implement. At the same time, compared with the ILP-based method, their time complexity and space complexity are lower, which is more friendly to the systems that require timeliness. However, these strategies often lead to local optimum results, which may not be the best result.

(2) ILP-based Strategies. ILP-based strategies convert the fair recommendation into an allocation problem using the user recommendation list as a container. They regard user satisfaction and fairness as objective and constraints respectively (or vice versa) and solve it through integer linear programming algorithms.

In Reference [7], relevance-based rankings are perturbed to improve fairness, which might lead to lower ranking quality. To address the tradeoff, they formulate the constrained optimization problems by minimizing unfairness and being subject to constraints on quality. The optimization problem is solved as an integer linear program.

Reference [91] also formulates an ILP problem. For customer u logging in at time t , n decision variables $X_{u,i}$ are introduced, which is set to 1 if i is recommended to u and to 0 otherwise. Assuming a specific customer u logs into the platform at time t , the problem is formulated as follows:

$$\begin{aligned} \min \quad & \sum_{i \in I} \left| g_i^t + \frac{X_{u,i}}{k} - \bar{D}_i^t \right| \\ \text{subject to } & X_{u,i} \in \{0, 1\}, \quad (b) \\ & \sum_{i \in I} X_{u,i} = k, \quad (c) \\ & \sum_{i \in I} X_{u,i} \cdot v_{u,i} \geq \theta_i \cdot \sum_{i \in I_u^{ori}} v_{u,i}, \quad (d) \end{aligned}$$

where g_i^t is the exposure of i at t , $1/k$ is the attention to i from u , and $\bar{D}_i^t$ is the target exposure proportion for i in step t . Here constraint (b) ensures the variables binary are kept. Constraint (c) ensures exactly k items are selected. A minimum customer utility is guaranteed by constraint (d).

Reference [69] provides a framework that can generate fairness-aware recommendation lists based on a fairness-constrained re-ranking method. The original recommendation lists are used as the baseline, and a re-ranking algorithm is applied to maximize the sum of the preference scores under the fairness constraint to generate new fair *top-k* recommendation lists. The objective function can be interpreted as that selecting exactly k items out of the baseline *top-N* list of each user so that the objective function can be maximized. The optimization problem here can be solved as a 0–1 integer programming problem, and they find feasible solutions to this NP-complete problem through fast heuristics.

The fairness-aware algorithm in Reference [36] considers both historical user interactions and the diversity of generated explainable paths. They introduce two fairness scores: the path score and the diversity score. The path score weights the quality of paths. The diversity score weights the diversity of paths. By aggregating the personalization and diversity scores, the fairness score can be calculated. The authors create a *top-k* recommendation list with items ranked in descending order by the optimized recommendation score and the fairness score. The optimization can be cast as a 0-1 integer programming optimization problem.

The ILP-based strategies require a lot of space and time to find the global optimal solution of the objective function, and the performance of the result is better than the heuristic algorithm. But as the number of individuals calculated by the algorithm increases, its solution set space and time consumption will increase exponentially, which becomes the bottleneck of the application of ILP-based strategies. Therefore, these algorithms are more suitable for small-scale platforms or offline recommendation scenarios that do not have high requirements for timeliness. At the same time, the authors often give an alternative heuristic algorithm or narrow the solution set space through pre-filtering to solve these problems.

4.4 Statistics of Fairness Assurance Strategies

We organize papers with different types of algorithms in recent years and briefly introduced each algorithm in Table 4. It can be seen that the in-processing algorithms are the mainstream of the fair

Table 4. Fairness Assurance Strategies

Reference	year	Stage	Type	Brief Introduction
[98]	2019	Pre-processing	Massaging the dataset	Augment the original data by generating artificial users (or items)
[12]	2021	Pre-processing and In-processing	Sampling and Regularization Term	A treatment combining observation upsampling and loss regularization
[11]	2021	Pre-processing and In-processing	Massaging the dataset	A combination of sampling strategy and regularized loss
[16]	2018	In-processing	Reweighting the Neighborhoods	Variants of SLIM
[149]	2018	In-processing	Debiasing Latent Factors or Embeddings	Isolating and extracting the sensitive information from the latent factor matrices
[83]	2018	In-processing	Multiple Objective Optimization	Jointly optimize for fairness and relevance
[146]	2018	In-processing	Multiple Objective Optimization	Combinatorial optimization based on the fair-efficiency function
[78]	2019	In-processing	Miscellaneous	Allocate applicant with the highest skill matching score
[6]	2019	In-processing	Regularization Term for Unfairness	Pairwise regularization to improve fairness
[73]	2019	In-processing	Multiple Objective Optimization	A two-step algorithm that theoretically guarantees the Pareto efficiency
[134]	2020	In-processing	Miscellaneous	A recommendation scheme combining a probabilistic model with coalition game strategy
[9]	2020	In-processing	Multiple Objective Optimization	A method manages the inherent loss of accuracy when fairness is increased
[85]	2020	In-processing	Regularization Term for Unfairness	Creating an error term pushes the items from the underexposed groups upwards in the ranking
[117]	2020	In-processing	Regularization Term for Unfairness	Regularizing the correlation between prediction errors and the distribution of market segments
[124]	2021	In-processing	Debiasing Latent Factors or Embeddings	A decomposed adversarial learning method with orthogonality regularization
[144]	2021	In-processing	Debiasing Latent Factors or Embeddings	A contrastive learning framework for debiased deep candidate generation
[30]	2021	In-processing	Debiasing Latent Factors or Embeddings	A joint framework encapsulating the GNN model utility and individual fairness
[121]	2021	In-processing	Miscellaneous	A framework performing multi-task training according to the causal graph
[70]	2021	In-processing	Debiasing Latent Factors or Embeddings	A model to generate feature independent user embeddings through adversarial learning
[39]	2021	In-processing	Miscellaneous	Modeling interactions as a Markov Decision Process and constraining fairness at each iteration
[53]	2021	In-processing	Debiasing Latent Factors or Embeddings	A deep neural network on big implicit feedback data for non-sensitive items
[127]	2021	In-processing	Debiasing Latent Factors or Embeddings	A fair graph based recommendation model for ensuring fairness
[40]	2022	In-processing	Debiasing Latent Factors or Embeddings	Generating feature-based explanations for the given black-box recommendation model
[94]	2022	In-processing	Debiasing Latent Factors or Embeddings	Learning provider fair news representations and provider-fair user representations
[41]	2022	In-processing	Debiasing Latent Factors or Embeddings	A fairness-aware recommendation framework using multi-objective reinforcement learning
[68]	2022	In-processing	Miscellaneous	An GANs-based solution consisting of a ranker and a controller
[122]	2022	In-processing	Debiasing Latent Factors or Embeddings	A fair meta-learning framework to enable fair adversarial learning in recommender system
[114]	2023	In-processing	Debiasing Latent Factors or Embeddings	A cross-domain recommender with fairness-aware objective and IF-based rewighting
[71]	2023	In-processing	Debiasing Latent Factors or Embeddings	A two-step framework from a learned fair model of the source users to the target users based on adversarial learning
[138]	2017	Post-processing	Generative Strategies	A greedy algorithm adding the item that covers the largest number of non-covered users each time
[7]	2018	Post-processing	ILP-based Strategies	An ILP-based fair ranking mechanism
[56]	2018	Post-processing	Generative Strategies	A heuristic algorithm selecting the item with the largest weighted combination of relevance and fairness
[42]	2019	Post-processing	Generative Strategies	Heuristic algorithms selecting items that will violate the ratio constraint
[112]	2019	Post-processing	ILP-based Strategies	Minimize income inequality while distributing customer waiting time fairly between the customers
[104]	2019	Post-processing	Heuristic Re-ranking Strategies	A re-ranking algorithm that optimizes for a desired tradeoff between utility and fairness
[76]	2019	Post-processing	Heuristic Re-ranking Strategies	A personalized re-ranking algorithm with a diversity tolerance term
[91]	2020	Post-processing	ILP-based Strategies	An ILP-based incremental update mechanism
[110]	2020	Post-processing	Heuristic Re-ranking Strategies	An aggregation method that is a weighted combination of the average and the least misery
[32]	2020	Post-processing	Heuristic Re-ranking Strategies	Post-processing the recommendation matrix with minimum impact on the utility of users
[130]	2020	Post-processing	Heuristic Re-ranking Strategies	A heuristic algorithm that guarantee the long-term fairness
[57]	2020	Post-processing	Generative Strategies	A greedy algorithm for fairness across the group members in a rank-sensitive way
[36]	2020	Post-processing	ILP-based Strategies	Optimizing recommendation score and fairness score through 0-1 integer programming
[90]	2020	Post-processing	Heuristic Re-ranking Strategies	A greedy algorithm providing a minimum guarantee on the exposure of the producers
[148]	2021	Post-processing	Heuristic Re-ranking Strategies	A model-agnostic False Positive Correction method
[150]	2021	Post-processing	Generative Strategies	A learnable framework consisting of a joint-learning generative method and a score scaling model
[131]	2021	Post-processing	Heuristic Re-ranking Strategies	A two-sided recommendation model for both customers and providers
[46]	2021	Post-processing	Heuristic Re-ranking Strategies	A re-ranking policy that maximizes utility while respecting the fairness criteria
[69]	2021	Post-processing	ILP-based Strategies	Maximizing the sum of the preference scores under the fairness constraint
[106]	2022	Post-processing	Generative Strategies	Enumerating all fair packages to allow users to select their favorite packages
[86]	2022	Post-processing	ILP-based Strategies	An re-ranking approach that integrates fairness constraints in a joint objective framework

recommendation algorithm, and the post-processing algorithms are also popular. But the research on pre-processing algorithms seems to have stalled.

An obvious research trend can be seen in the statistical results of in-processing algorithms, that is, over time, the research focus gradually shifts from traditional machine learning algorithms to new neural network structures and deep learning algorithms. Among post-processing algorithms, heuristic algorithms have received more attention due to their low time complexity and ease of application.

There is also an obvious trend that the algorithms and researches to ensure fairness in recommender systems are increasing year by year, which shows that the issue of fairness in recommender systems is gradually attracting more and more attention.

4.5 Pros and Cons of Different Strategies and Selection Guidelines

We summarize the pros and cons of different strategies in Table 5. Since the detailed designs of the algorithms are different, we only compare strategies qualitatively.

The pre-processing strategies improve fairness by eliminating bias in the data, which makes them suitable for almost all kinds of recommenders without modifying the subsequent recommendation models. But it is worth noting that the pre-processing algorithms often need to obtain the distribution of some features in the original data, as well as some macroscopic statistics. These requirements are relatively easy to satisfy in an offline manner but may consume a lot of computing time and spaces for doing it online. At the same time, theoretical guarantees are not provided for the performance under different scenarios. The modifications to the original data also bring the concerns about motivations and the adverse effects. Since there are still very complex algorithm

stages after processing the data, it is difficult for us to predict the impact of pre-processing. Although there will be some deviations in data sampling, how to ensure that the processed data are completely in line with the real situation instead of adding the influence of subjective factors is also challenging. Since only some rule-based processing is usually performed on the raw data, the time complexity and space complexity of the pre-processing strategies are generally low.

Most algorithms proposed in recent years fall into the category of in-processing strategies. Their effectiveness are often proved by theoretical analysis approaches or experiments. People can select algorithms that can satisfy their requirements from a number of algorithms. However, the in-processing strategies need to modify the original recommender system, which requires users to have a full understanding of the original algorithm and their own fairness requirements before they can choose a suitable algorithm. Since in-processing strategies are usually bundled with machine learning algorithms, their time and space complexity are relatively high. Adding a fairness regularization term is a very intuitive method. However, an inappropriate fairness regularization term may cause the model to fail to converge or impair other performance metrics. Therefore, when using this method, the weights of the fairness regularization term and the other terms of the loss function should be set carefully. Moreover, if the fair recommendation problem is converted into a multi-objective optimization problem, then we can use multi-objective optimization algorithms to solve it. However, it may take a lot of time to find the Pareto front. With the development of representation learning, mapping users and items to the latent space for learning has become the mainstream solution, which converts some difficult-to-use data into vectors that can be used by more algorithms. However, the interpretability of this method is poor, and the impact on the results is also difficult to predict, leading to it often requiring a lot of practice to obtain good results. However, a neighborhood-based approach may be more suitable for people who are not familiar with deep learning. At the same time, it is worth noting that an algorithm often does not simply belong to one strategy type but a combination of multiple ones. For example, it is a common practice to add a fairness regularization term to the loss function in the learning process of fair embedding.

Since there is no need to change the original recommendation model, the post-processing strategies also have the advantage of strong adaptability. Because the original results are processed directly, the side effects can be controlled through some constraints. At the same time, heuristic re-ranking algorithms as well as generative strategies generally have the advantages of low time and space complexity, which are preferred by large-scale systems. In contrast, when dealing with a huge dataset, the ILP-based method requires a large combination space, and optimization needs to be executed in each round. Thus, the time and space complexity of the ILP-based algorithm are very high.

In conclusion, post-processing strategies have a lower learning curve and easy to implement. It can be applied without understanding the details of the original recommender algorithms and feature distributions in the data. However, when fairness requirements are explicit and the original recommender algorithms are well understood, in-processing strategies can be tried, because they have theoretical guarantees if properly designed. For the adoption of pre-processing strategies, we should be cautious, since we do not know what kind of side effects will be introduced.

It should be noted that we need to clarify the notions of fairness adopted and the speciality of the applications before making a choice among different strategies.

5 FAIRNESS IN DIFFERENT RECOMMENDATION TASKS

There are different recommendation tasks in practice. Generally, recommender system can be divided into non-personal ones and personal ones. For non-personal recommendation, the items to be recommended are decided based on popularity, timeliness, cost, or quality. The selection of

Table 5. Pros and Cons

Category	Applicability	Domain knowledge dependency	Time complexity	Space complexity	Available algorithm number
Pre-processing	high	a certain amount	low	low	few
In-processing	low	much	high	high	abundant
Post-processing (Generative Strategies)	moderate	a little	low	low	a number of
Post-processing (Heuristic Re-ranking Strategies)	high	little	low	low	a number of
Post-processing (ILP-based Re-ranking Strategies)	high	little	high	high	a few

items may not be fair among people from different groups. The main challenge here is the data bias problem if only user fairness is considered. However, when provider fairness is also considered, it becomes a multiple objective optimization problem.

Currently, most research focuses on the personal recommendation problem, which can further be divided into different task types. Rating prediction and ranking are two main recommendation tasks. We have listed many fairness metrics for them in Section 3.4.

There are many other recommendation tasks including group recommendation, package recommendation [75], cross-domain recommendation [58], social-based recommendation [113], sequential/session-based recommendation [89], conversational recommendation [23, 141], and so on.

Group recommendation refers to recommending items to a group of users, which means the preferences of the entire group need to be considered. Starting from the least misery principle, Reference [103] defines the fairness of the group as the utility of the minimum member utility. References [57, 110] consider fairness within the group and hope that the results of the recommendation can balance the interests of all group members. It can be seen that the current group recommendation mainly considers fairness within the group, but in fact, there will be unfair phenomena between groups due to differences in group size and other differences, which is also a direction worth studying in the future.

Package recommendation [75] refers to recommending a group of items to users. In Reference [108], the fairness problem in package-to-group recommendations is studied. Specifically, the recommendation is fair in the sense that every group member is satisfied by a sufficient number of items in the package. For individual package recommendation, there is no study on the fairness problem yet. When a package of items is recommended to a user, we believe the unfairness among either users or providers becomes more serious, because the package is generated by considering the inherent relationships among items, which may reinforce the bias in item selections.

Cross-domain recommendation [10, 58, 142] leverages feedback or ratings from multiple domains to improve recommendation accuracy in a collective manner. When we learn knowledge from other domains, the bias in these domains can also have an impact. How to avoid the impact of bias while learning useful knowledge from other domains is an interesting topic. So far, there is no research on the fairness issue in cross-domain recommendation.

Social-based recommendation [113] uses social information to help mitigate the data sparsity and cold start problems. In social networks, the phenomena of homophily and influence explain the fact that friends tend to be similar. Similarly to cross-domain recommendation, how to avoid biased information learned from friends is also a research topic in social-based recommendation. In Reference [105], it is pointed out that social-based recommender system tend to make friends appear more similar compared to non-socially aware techniques so that some users' individual preferences are suppressed more compared to other users in their social circle. An approach to identify who they are and techniques to protect them have been designed.

Session-based recommendation [118] is usually based on the user's anonymous click or access chain on the web page or a series of user purchase records to recommend the next item to the user. However, sequential item dependency relationships may reinforce unfairness both in users and providers. And because less information is available, more severe polarization may appear in the

results. At the same time, the dynamic changes of user preferences and the multiple interactions between users and the system of session recommendation are both new research directions for the fairness of recommendation systems. Reference [97] proposes a fair session-based recommendation system, using Gated Graph Neural Network as graph embedding and DeepWalk as node embedding. Reference [66] not only aims to accurately predict the next items but also requires that the recommended items lead to **Interaction Fairness (IF)**. Better IF means that recommended items tend to be equally interacted by users of different protected attribute groups.

6 DATASETS

Since the research on fairness in recommender systems is also carried out under the framework of recommender systems, therefore, the most of the datasets used in fairness research in recommender systems are the same as the datasets used in recommender system study.

The MovieLens dataset [48] is the most popular dataset in the research on recommender systems. This dataset contains a set of movie ratings from the MovieLens website, a movie recommendation service. This dataset was collected and maintained by GroupLens, a research group at the University of Minnesota. There are five versions included: “25m,” “latest-small,” “100k,” “1m,” “20m.” In all datasets, the movies data and ratings data are joined on “movieId.” The 1m dataset and 100k dataset contain demographic data in addition to movie and rating data, which are also two of the most commonly used datasets.

The Amazon Review dataset [80] contains product reviews and metadata from Amazon, including 82.83 million reviews spanning May 1996–July 2014. This dataset includes reviews (ratings, text, helpfulness votes), product metadata (descriptions, category information, price, brand, and image features), and links (also viewed or bought graphs).

The Google local datasets [49] contain reviews about businesses from Google Local (Google Maps), as well as the geographic information for each business.

There are also representative datasets in some sub-fields, such as the Last.fm dataset [17] and the Million Song dataset [81] in music recommendation; the COCO Course Collection [28] in course recommendation; the Andriod Market dataset [146] in mobile service recommendation; Kiva’s dataset [76] in loan recommendation; the Ciao dataset [115]; Facebook, Twitter, and Google Plus data [64] in social networks; and so on. There are also a few studies to test the effectiveness of the algorithms in the real online systems through online A/B tests [42, 144], and some studies test the performance of the algorithm on the synthetic datasets [119, 130].

For the convenience of comparison, the information of available datasets used in research on fairness in recommender systems is listed in Table 6. The first column of the table is the name of the datasets and the source article or URL. The second column is the application domain of the datasets. The third column contains the number of users, items, and interactions between users and items in the dataset. The fourth column represents the user-side and provider-side sensitive attributes in the dataset as well as some additional information. The next column shows which class fairness metrics the dataset is suitable for, and we classify the metrics in terms of *Fairness for Whom* and *Demographic Units* in terms of the framework we propose. The last column shows the papers using this dataset.

¹<https://tianchi.aliyun.com/dataset/dataDetail?dataId=56>.

²<https://tianchi.aliyun.com/competition/entrance/231721/introduction>.

³<https://sites.google.com/site/yangdingqi/home/foursquare-dataset>.

⁴<https://www.kaggle.com/mrmorj/insurance-recommendation>.

⁵<https://www.upf.edu/web/mtg/kgrec>.

⁶<https://msnews.github.io/index.html>.

⁷<http://www.yelp.com/dataset> challenge.

Table 6. Dataset Information

Datasets	Domain	Users	Items	Interactions	User-side Attributes	Provider-side Attributes	Other Information	Individual	Group		Used by
									User	Provider	
Address [44]	news	13.5K	744	116.3K	position, device type	category, word count	timestamps	✓	✓	✓	[121]
AlIEC [132]	E-Commerce	60.0K	88.0K	2.1M	gender, age	category, brand	timestamps	✓	✓	✓	[132]
Amazon Review [80]	product review	30.0M	9.4M	82.8M	gender	category, price	timestamps	✓	✓	✓	[36, 69, 86, 91, 96, 131]
Amazon Electronics [117]	product review	1.2M	9.6K	1.3M	gender	—	timestamps	✓	✓	—	[117]
Book-Crossing [151]	book	278.9K	271.4K	1.1M	age, location	year, publisher	—	✓	✓	✓	[66, 86, 96, 122]
Ciao [115]	product review	12.4K	106.8K	484.1K	—	category	—	✓	—	✓	[41, 134, 148]
CIKM2019 E-Comm AF [147]	E-Commerce	12.4K	106.8K	484.1K	gender, age, purchasing power	category	—	✓	✓	✓	[153]
COCO [28]	course	37.7K	30.4K	617.6K	—	category	timestamps	✓	—	✓	[11, 12]
Ctrip Flight [131]	air ticket	3.8K	6.0K	25.2K	—	airline, class, price	—	✓	—	✓	[131]
EC-REC [73]	E-Commerce	—	—	7.0M	—	price, the labels of impression	—	✓	—	✓	[73]
Epinions [115]	product review	22.2K	296.3K	922.3K	—	category	—	✓	—	✓	[86, 96, 134]
Foursquare ² [135]	check-in	266.9K	3.7M	33.3M	—	location, category	timestamps	✓	—	✓	[86, 96]
Globo [26]	news	158.3K	12.0K	2.5M	position	category, word count	timestamps	✓	✓	✓	[121]
Google Local [49]	business review	4.6M	3.1M	11.4M	—	category, price, location	timestamps	✓	—	✓	[90, 91, 131]
Gowalla [22]	check-in	20.9K	41.0K	1.0M	—	location	timestamps	✓	—	✓	[86, 96, 121]
Insurance ³	insurance	1.2K	21	5.4K	gender, age, occupation, marital status	—	—	✓	✓	—	[70]
KGRec-music dataset ⁴	music	5.2K	8.6K	751.5K	—	category	—	✓	—	✓	[57]
Last.FM 1K [65]	music	992	177.0K	19.2M	gender, age	—	—	✓	✓	—	[34, 35, 86, 90, 96]
Last.FM 360K [65]	music	359.3K	160.2K	17.6M	gender, age	—	—	✓	✓	—	[29, 34, 35, 71, 127]
MIND ⁵ [125]	news	1.0M	130.4K	5.0M	—	category	timestamps	✓	—	✓	[94]
ModCloth [117]	product review	44.8K	1.0K	99.9K	body shape	size	timestamps	✓	✓	✓	[117]
MovieLens 100K [48]	movie	943	1682	100.0K	gender, age, occupation	category	timestamps	✓	✓	✓	[39, 41, 63, 86, 122, 126, 149]
MovieLens 10M [48]	movie	69.9K	10.7K	10.0M	—	category	timestamps	✓	—	✓	[12, 121]
MovieLens 1M [48]	movie	6.0K	3.7K	1.0M	gender, age, occupation	category	timestamps	✓	✓	✓	[9, 16, 34, 98, 103, 138] [11, 32, 57, 70, 148, 150] [29, 39, 53, 71, 96, 127] [66, 106, 122, 126]
MovieLens 20M [48]	movie	138.5K	27.3K	20.0M	—	category	timestamps	✓	—	✓	[85, 104, 110, 130]
OK-article [139]	article	1.3M	220.1K	46.1M	—	category	—	✓	—	✓	[114]
OK-video [139]	video	5.0M	3.8M	142.3M	gender, age	—	—	✓	—	✓	[114]
XING [1]	job	1.5M	1.3M	314.5M	career level, experience years, position	type of employment, location	timestamps	✓	✓	✓	[97, 150]
Yelp ⁶	business review	2.0M	150.3K	7.0M	—	category, location	timestamps	✓	—	✓	[40, 108, 121, 130]

Recommender system datasets usually contain interaction information between users and items, which is sufficient for studying *individual fairness* in recommender systems. However, group fairness needs to group users or providers, so only datasets that contain sensitive attributes of users or providers can be used for research on *group fairness*. It is worth noting that the group here refers to the demographic groups based on sensitive attributes. If the groups are artificial groups, then it is not restricted by sensitive information. For single-round and long-term fairness, it is not necessary to pay attention to whether the dataset contains time information, and it can be achieved by artificially increasing the number of recommended rounds. In addition, in the datasets with timestamps, the users' interactions can be sorted by time to form sessions, and research on the fairness of sequential or session-based recommendations can be carried out.

7 CHALLENGES AND FUTURE WORK

The studies on fairness in recommender systems are facing several challenges, which also implies some potential research topics.

7.1 Lack of Applicability Criteria of Fairness Notions

Fairness is an abstract concept, which means it cannot be applied directly. To ensure that fairness measurement and assurance strategies can be implemented, a couple of fairness notions are further proposed for machine learning, and some of these have been adopted for recommender systems.

Although fairness notions make the concept of fairness more tangible, they are adopted in an ad hoc manner in different studies. Specifically, there is no comprehensive research on the applicability of different fairness notions in recommender systems. As a starting point, in Reference [24], the applicability of three notions is studied. It is necessary to define the criteria for the applicability of fairness notions. Moreover, decision guidelines are needed for practitioners and policy makers to select the appropriate ones from the large catalogue of fairness notions.

7.2 Lack of Unified Fairness Metrics

The fairness metrics proposed are diverse. Even for the same fairness notion, the fairness metrics to implement it in studies are quite different.

This can be attributed to the fact that the evaluation metrics of recommender systems are inherently multi-dimensional and change with time. In addition to basic metrics such as accuracy,

people gradually have paid more attention to other metrics such as novelty and diversity of recommendation results. For example, dining recommendations should try to recommend healthier foods when considering users' preference. Moreover, a user's preferences for different aspects of recommendation quality change dynamically. All of these make the evaluation of recommendation systems a very complicated task. When the fairness metric is considered, it further increases the complexity of the problem.

When different metrics are applied in different studies, the performance of various approach is not comparable. Although it is not possible to provide a unique fairness metrics, it may still be possible to provide a set of metrics for a specific situation so that a consistent basis can be formed for algorithm design and performance comparisons.

7.3 The Dilemma of User Classification

Under the notion "fairness through awareness," which is widely adopted in many studies, it requires the definition of similar individuals with respect to a similarity metric, i.e., to classify users in terms a similarity metric. However, to propose a similarity metric that is proper and meaningful in practice is not easy.

Moreover, to classify users into different categories may imply unfairness to some degree. For example, for heavy users, due to the sufficient data available, it is relatively simpler to ensure their recommendation quality. But for users who access the system occasionally, their data are not enough so that recommendation quality cannot be ensured. To achieve fairness in quality for all individuals is not reasonable. Therefore, we can divide them into groups according to the amount of their behavior data. However, this classification implies consent to the unfairness of recommendation quality between users with different amounts of data.

As preliminary research, in Reference [107], suggestions are proposed to increase fairness with regard to user-side diversity. The idea is to contextualize user diversity (i.e., user classification) by mapping structural differences (inequalities) between users, which materialize as different experiences of privilege and oppression.

7.4 Balancing Fairness and Quality Requirements

Fair recommendation algorithms need to solve the problem of recommendation quality loss caused by improving fairness. This seems to be an inevitable problem when introducing fairness measures into recommender systems. At the same time, the problem is also very complex. First, the measurement of recommendation quality is multi-dimensional, and the introduction of fairness will further increase the dimensions of problem consideration. Second, how to define a persuasive tradeoff criterion is also a complicated issue.

Currently, a global weighting schema is used to balance the requirement of fairness and quality measurement is a typical approach. However, different people may have different sensibilities to quality loss. To differentiate the balance requirements among people is an interesting topic. It is possible to learn this difference if more data are available.

7.5 Balancing Long-term Fairness and Fairness in a Single-round Recommendation

Some algorithms try to solve the fairness problem for single-round recommendations, and others try to solve the long-term fairness problem. Intuitively, ensuring the fairness of a single-round recommendation can also guarantee long-term fairness, but in actual situations, the algorithm cannot guarantee the absolute fairness of a single-round recommendation, which will cause unfairness to be accumulated in the long-term recommendation process. On the contrary, guaranteeing long-term fairness cannot ensure the fairness of each single-round recommendation. If handled improperly, then it will cause users to experience ups and downs in recommendation quality.

Another problem relating to long-term fairness is that users have different access times. Therefore, only the average fairness over multiple round recommendations is meaningful. However, compared with users with more access times, the variance of the average fairness metrics of users with less access times is naturally larger when the same algorithm is applied.

7.6 Multilateral Fairness

Generally, fairness is defined and measured among individuals or groups of the same type. However, on a more abstract level, if we treat each party in the ecosystem as a whole, then we can also define fairness to maintain the equality of their utilities, which we refer to as the multilateral fairness problem.

More specifically, to ensure the utility level of the platform and maintain recommendation quality, this can also be modeled as a multilateral fairness problem. The platform itself is also an important stakeholder in the recommender system. The interests of the platform are more related to profits, reputation, quantity of users access, and number of active users. Although improving recommendation quality benefits users that also enhances their reputation and increases their number of active users, it may not be necessary to increase profits. The same situation occurs when the platform has different pricing rates for different providers. Therefore, to maintain a profit level and to ensure the interests of users or providers can be contradictory to some degree. This results in a multilateral game among the platform, users, and providers that is expected to produce fair results.

7.7 Personalized Fairness Requirements

With the popularization of the concept of fairness in recommender systems, users may have their own personalized fairness requirements. Moreover, users can have different fairness requirements in different scenarios. This leads to a recommender system should assure different fairness notions at the same time. However, the fairness notions are not consistent or even conflict with each other.

7.8 Computation Complexity

The fairness measurement is usually a global indicator, each calculation needs to traverse the full-scale data, and some algorithms need to be processed repeatedly and the measurement is updated after the processing before proceeding to the next step. Designing some indexes with low computational complexity or incremental indexes will be interesting research directions. In addition, redesigning some incremental algorithms or locally updating data can also effectively solve this type of problem.

7.9 Privacy Issues

When dealing with fairness among users, algorithms usually need to access various attributes of users. But as users pay increasing attention to personal privacy, many users choose to be anonymous or hide some protected parameters when using a recommendation system. How to ensure the fairness of the recommender system when part of the information is missing, especially for the group fairness problem, which is more dependent on protected parameters and will be a very meaningful study.

For provider fairness, some fairness assurance strategies do not necessarily rely on privacy data. However, to calculate user fairness the users' attributes need to be known. Privacy computing [67] can be a solution to this problem.

7.10 Other Issues

In addition to solving the above problems, we believe that the issue of fairness in recommender systems has some other interesting research directions. For example, we can combine the

state-of-the-art machine learning algorithms with fairness to further improve the effectiveness of the system. We can also consider the exchange of utility among items from different domains so that a loss in recommendation quality in one item domain can be compensated by the better quality in another item domain. This can achieve a fair state of items from multiple domains as a whole. Last, building some standard datasets for the fairness of recommendation systems is also very important for the advancement of this area.

8 CONCLUSIONS

In this article, we presented a review on the fairness issue in recommender system. Although there are several surveys on the fairness issue in machine learning, the work on machine learning is not sufficient to address the fairness issue in recommender systems, because this problem is complex, since the objectives of recommender systems are diversified. Even though there are several reviews on the fairness issue in recommender systems, they are not sufficient in terms of the scope and depth. In particular, they neglect the links between fairness issues and the context of different recommender systems. Therefore, we propose a more comprehensive framework to classify fairness metrics from four dimensions. We also provide an in-depth analysis on the fairness assurance strategies and the datasets used by the algorithms. Finally, we present some challenges and discuss future work in terms of these challenges. Since fairness is a subjective term, the diversity of fairness definitions in research cannot be avoided. On the one hand, the diversity of fairness definitions is the source of the research value. On the other hand, to establish some consensus is necessary to promote study in this area. It is expected that in the future, more definitions will evolve and, at the same time, efforts will be made to provide a unified framework.

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